

BALU BHASURAN

Research Faculty I, School of Information, Florida State University

PROFESSIONAL SUMMARY

Experienced researcher and educator specializing in clinical informatics, biomedical text mining, and machine learning applications in healthcare. My current work centers on Clinical NLP, Machine Learning, and Generative AI. With over 12 years of experience, I have developed various classification and prediction models using biomedical and clinical data. My interests include machine learning based prediction models, knowledge graphs, causal inference, literature-based discovery, network visualization, and ensuring fairness and explainability in machine learning models.

EMPLOYMENT HISTORY

RESEARCH FACULTY I, VISITING ASSISTANT PROFESSOR, POSTDOCTORAL RESEARCHER **Jul 2023 - Present**
Florida State University Tallahassee, FL

LIS 5916 – Issues in Information Studies: Natural Language Processing and Text Mining for Information Technology Professionals

Role: Instructor (Spring 2026, Spring 2025)

LIS 4761 and LIS 5765 – Data Mining and Analytics: Role: Teaching assistant

Supervised investigation of LLMs response to patient-centric lab test result-related questions.

Investigation into Retrieval augmented Generation (RAG) based LLM systems for patient-centric questions

Development of a machine learning prediction model for pediatric transplantation rejection and HIV medication adherence and disease prediction using EHR and transplant registry data

Investigation of LLMs responses to clinical case report-based differential diagnosis, causal reasoning using lab test data.

Investigation of the role of seasonality in lab test results of Alzheimer's and Dementia patients using EHR data.

POSTDOCTORAL RESEARCHER **Jan 2020 - Jul 2023**
University of California, San Francisco, San Francisco, CA

Orchestrated training and clinical testing of early diagnosis model for rare disease using electronic health records from UCSF and UCLA

Developed NLP algorithms for NASH patient identification and adverse drug event extraction from clinical notes.

Invented a text classification algorithm for Mayo endoscopic sub-scores using clinical notes

JUNIOR AND SENIOR RESEARCH FELLOW **Jul 2014 - Jun 2018**
DRDO-BU Center for Life Sciences India

Developed and utilized machine learning models for High-Altitude Diseases research

Innovated in text mining techniques for biomedical data

Authored three webservers for efficient information extraction, securing Indian Copyright

EDUCATION

PHD, COMPUTATIONAL BIOLOGY **Sep 2014 - Dec 2020**
Bharathiar University, India

Thesis: Biomedical Text Mining Approaches: Applications in Disease Entity Recognition, Gene-Disease Association Extraction and Knowledge Discovery

Topics: Information Extraction, Biomedical Informatics

Thesis: Data Mining to predict efficient allocations in electrical grids

Relevant Coursework: Data structures, Algorithm Analysis and Design

SKILLS

Natural Language Processing, Machine Learning, Biomedical Informatics, Clinical Data Analysis, Predictive Analytics, Algorithm Development, Curriculum Development, Academic Research, Course Design, Student Assessment, Team Collaboration, Communication.

LINKS

Personal Website: balubhasuran.github.io, GitHub: github.com, Google Scholar: scholar.google.co.in, LinkedIn: www.linkedin.com.

ADDITIONAL INFORMATION

GRANTS

DEVELOPING A MULTI-AGENT AI SYSTEM FOR EXPLAINING LAB RESULTS TO OLDER ADULTS". NATIONAL A2 PILOT AWARDS. NATIONAL INSTITUTE ON AGING. \$280,800 (10/2025 – 3/2027)- Co PI

FINE-TUNING OPEN-SOURCE LARGE LANGUAGE MODELS TO SUPPORT REAL-WORLD PATIENT–CLINICIAN COMMUNICATION IN LABORATORY TEST RESULTS INTERPRETATION, NATIONAL ARTIFICIAL INTELLIGENCE RESEARCH RESOURCE (NAIRR) PILOT, NATIONAL SCIENCE FOUNDATION (NSF) (NAIRR250293), 2,000 H100 GPU HOURS (PURDUE ANVIL AI) 2025-12-15 TO 2026-12-14- Co PI

PATENTS

Methods for improving the diagnosis of rare diseases using electronic health records data and systems for same

Inventor: Vivek RUDRAPATNA, Balu BHASURAN, WO2024049873A1 WIPO (PCT)

Identification of Medical Intervention Related Adverse Events from Clinical Notes

Inventor: Vivek Rudrapatna, Rebecca Racz, Mahalakshmi Parakala, Madhumita Sushil, Ludwig Dana, Balu Bhasuran, Anna Silverman, James Buchanan

PCT/US2024/028389, U.S. provisional patent application serial no. 63/465,382 filed May 10, 2023

SOFTWARE COPYRIGHTS

- Indian Copy Right, Computer Software, SW-15957/2023, D-NER: Disease Name Recognizer from Literature
- Indian Copy Right, Computer Software, SW-15958/2023, DisGeReExt: Disease Gene Relation Extractor from Literature
- Indian Copy Right, Computer Software, SW-15974/2023, GD Miner: Gene-Disease Association Mining From Literature

AWARDS

- American Transplant Congress Pediatric Poster Award Winner, 2024
- Best paper award, Data Mining, International Conference on Innovative Computing and Communication (ICICC-2018)
- Best poster award, Text Mining, International Symposium on Computational Biology and Bioinformatics (BioIndica- 2016)
- Best poster award, Text Mining, Fifth Edition of National Workshop on Computer Vision, Image Processing Techniques and Data Analytics, 2015

CERTIFICATIONS

- Microsoft Certified Professional (MCP) in Microsoft .NET Framework Application Development Foundation
- Developing Enterprise Applications using VC#.Net, NIIT, New Delhi, India

- National Eligibility Test (UGC .NET) in Computer Science and Applications for Assistant Professor, Govt.of India

RESEARCH AND TRAVEL GRANTS

National Institute of Health

Jul '24 – Dec '24

Principal Data Scientist, Prediction of Health Outcomes and Adverse Events in Pediatric Organ Transplantation in Florida

Grant number R21LM013911

AHRQ

Mar'24-Feb'29

Principal Data Scientist, LabGenie, a web-based tool intended to improve patients' engagement in managing and acting upon their lab test results.

Merck

Aug '22 – Jul '23

Principal Data Scientist Patient trajectories and risk prediction in patients with nonalcoholic steatohepatitis disease. Grant number UL1TR001872

Alnylam

Jan '21 – Jul '22

Principal Data Scientist, Reducing diagnostic delays in Acute Hepatic Porphyria disease. Grant number: K99LM014099, UL1TR001872, UL1TR001881

US Food and Drug Administration (FDA)

Feb '21 – Aug '22

Collaborator, Improving adverse event detection related to biologic immunosuppressant use – a pilot study of the BERT deep learning model adapted to real-world clinical notes.

Defense Research and Development Organization (DRDO)

Data Mining and Text mining to Identify Enzymes that are potential candidates for high altitude diseases.

Jul '14-Jun' 18

MENTORSHIP AND THESIS COMMITTEE

Undergraduate Research Opportunity Program (UROP),

FSU

Aug'23 – Present

LLM evaluation in differential diagnosis, hallucination detection and lab test result-related questions

The Young Scholars Program (YSP),

FSU

Aug'24 – Present

LLM evaluation in lab test result-related questions and reasoning

Rotation and PhD students

June '19 – present

Deep Learning for MRI Based Alzheimer's Disease Dementia Classification

Comparative Effectiveness of Ustekinumab and Vedolizumab in TNF-exposed Pediatric Patients with Ulcerative Colitis

Web application development for visualization of lab test results

ROLES

Associate Editor, Clinical and Experimental Medicine, Springer, 2025-Current

Editorial Board Member, BMC Medical Informatics and Decision Making, Springer, 2025-Current

Editorial Board Member, Discover Artificial Intelligence, Springer, 2025-Current

Review Board Member, Graduate Studies Academic Regulations and Procedures, University of Nizwa 2023 – 2025

Guest Editor, Information, MDPI, 2025-2026

Transformative Technologies in Healthcare: Harnessing Machine Learning, Deep Learning and Large Language Models in Health Informatics

Guest Editor, Life, MDPI, 2026-2027

Transformative Technologies in Healthcare: Harnessing Machine Learning, Deep Learning and Large Language Models in Health Informatics

Publication Chair

The 12th IEEE International Conference on Healthcare Informatics (IEEE ICHI 2024), Orlando, Florida, USA, June 3rd-6th, 2024

Scientific Program Committee AMIA 2025 Clinical Informatics Conference

JOURNAL REVIEWING

1. The Lancet Digital Health, Lancet
2. eBioMedicine, Lancet
3. npj Digital Medicine, Nature
4. npj Health Systems, Nature
5. Scientific Data, Nature
6. Signal Transduction and Targeted Therapy, Nature
7. Journal of the American Medical Informatics Association (JMIA), Oxford
8. Bioinformatics, Oxford
9. Journal of Medical Internet Research (JMIR), JMIR
10. Artificial Intelligence in Medicine (AIIM), Elsevier
11. Journal of Biomedical Informatics (JBI), Elsevier
12. Computational Biology and Chemistry, Elsevier
13. IEEE Journal of Biomedical and Health Informatics (JBHI), IEEE
14. PLoS Digital Health, PLOS
15. BMC Medical Informatics and Decision Making, BioMed Central
16. NAR Genomics and Bioinformatics, Oxford
17. Biology Methods & Protocols, Oxford
18. European Heart Journal - Digital Health, Oxford
19. Bioinformatics Advances, Oxford
20. BMJ Health & Care Informatics, BMJ (British Medical Journal)
21. BMJ Open, BMJ (British Medical Journal)
22. Journal of Biomedical Semantics, Springer Nature
23. BMC Bioinformatics, BioMed Central
24. Journal of Healthcare Informatics Research (JHIR), Springer Nature
25. Scientific Reports, Springer Nature
26. Knowledge Based Systems, Elsevier
27. BioData Mining, Springer Nature
28. Gene, Elsevier
29. Smart Health, Elsevier
30. JMIR Formative Research, JMIR
31. JMIR Dermatology, JMIR
32. JMIR Medical Informatics, JMIR
33. IEEE Access, IEEE
34. IEEE Transactions on Artificial Intelligence, IEEE
35. IEEE Transactions on Computational Social Systems, IEEE
36. IEEE Transactions on Human-Machine Systems, IEEE
37. IEEE Transactions on Affective Computing, IEEE
38. IEEE Transactions on Knowledge and Data Engineering, IEEE
39. IEEE Transactions on Multimedia, IEEE
40. PLoS ONE, PLOS
41. BMC Cancer, BioMed Central
42. BMC Public Health, BioMed Central
43. BMC Gastroenterology, BioMed Central
44. BMC Cardiovascular Disorders, BioMed Central
45. BMC Women's Health, BioMed Central

46. PEC Innovation, Elsevier
47. Engineering Applications of Artificial Intelligence, Elsevier
48. Intelligence-Based Medicine, Elsevier
49. Journal of Biomedical Research, Elsevier
50. Array, Elsevier
51. Journal of Infection, Elsevier
52. Data in Brief, Elsevier
53. Alexandria Engineering Journal, Elsevier
54. Artificial Intelligence Review, Springer Nature
55. Evolutionary Intelligence, Springer Nature
56. Journal of Medical Systems, Springer Nature
57. In Silico Pharmacology, Springer Nature
58. Discover Applied Sciences, Springer Nature
59. Journal of Rare Diseases, Springer Nature
60. International Journal of Machine Learning and Cybernetics, Springer Nature
61. International Journal of Data Science and Analytics, Springer Nature
62. Medical Oncology (MO), Springer Nature
63. Discover Data, Springer Nature
64. Journal of Cannabis Research, BioMed Central
65. Clinical and Experimental Medicine, Springer Nature
66. Italian Journal of Animal Science, Springer Nature
67. Journal of Alzheimer's Diseases, SAGE
68. Health Informatics Journal, SAGE
69. Biotechnology and Applied Biochemistry, Wiley
70. Clinical and Translational Discovery, Wiley
71. MedComm, Wiley
72. Journal of Biomolecular Structure and Dynamics, Taylor & Francis
73. Clinical Epidemiology, Taylor & Francis
74. Frontiers in Medicine
75. Frontiers in Artificial Intelligence
76. Frontiers in Oncology
77. Frontiers in Psychiatry
78. Frontiers in Research Metric and Analytics
79. Frontiers in Big Data
80. Heliyon
81. Drug Research, Thieme
82. Applied Sciences, MDPI
83. Big Data and Cognitive Computing, MDPI
84. Mathematics, MDPI
85. Data, MDPI
86. Informatics, MDPI
87. Information, MDPI
88. AI, MDPI
89. Healthcare, MDPI
90. Future Internet, MDPI
91. Forecasting, MDPI
92. Electronics, MDPI
93. Machine Learning and Knowledge Extraction, MDPI
94. Computers, MDPI
95. Hospitals, MDPI

96. International Journal of Environmental Research and Public Health, MDPI
97. Journal of Eye Movement Research, MDPI
98. Journal of Applied Bioinformatics & Computational Biology, SciTechnol
99. Artificial Intelligence in Health, ACS
100. Computers, Materials & Continua, Tech Science Press
101. International Journal of Medical Sciences

CONFERENCE REVIEWING

1. 17th ACM Conference on Bioinformatics, Computational Biology, and Health Informatics (ACM BCB 2026)
2. The 14th IEEE International Conference on Healthcare Informatics (ICHI 2026)
3. American Medical Informatics Association, Clinical Informatics, 2026
4. The International Conference on Artificial Intelligence in Medicine (AIME), 2026
5. American Medical Informatics Association, Annual Symposium, 2025
6. American Medical Informatics Association, Clinical Informatics, 2025
7. AIME 2025: 23rd International Conference of AI in Medicine, 2025
8. The IEEE/ACM International Conference on Connected Health: Applications, Systems and Engineering Technologies (CHASE), 2025
9. American Medical Informatics Association, Annual Symposium, 2024
10. American Medical Informatics Association, Clinical Informatics, 2024
11. AIME 2024: 22nd International Conference of AI in Medicine, 2024
12. The IEEE/ACM International Conference on Connected Health: Applications, Systems and Engineering Technologies (CHASE), 2024
13. American Medical Informatics Association, Annual Symposium, 2023
14. 9th International Conference on Smart Computing and Communication (ICSCC 2023)

PEER-REVIEWED PUBLICATIONS

1. Qiao Jin, Nicholas Wan, Robert Leaman, Shubo Tian, Zhizheng Wang, Yifan Yang, Zifeng Wang, Guangzhi Xiong, Po-Ting Lai, Qingqing Zhu, Benjamin Hou Maame Sarfo-Gyamfi, Gongbo Zhang, Aidan Gilson, Balu Bhasuran, Zhe He, Aidong Zhang, Jimeng Sun, Chunhua Weng, Ronald Summers, Qingyu Chen, Yifan Peng, Zhiyong Lu. Entry-level guide to the use of large language models for medical research. *Nature Protocols*. 2026. In press.
2. Bhasuran B, Prosperi M, Hanna K, Petrilli J, Washington CJ, He Z. Evaluation of causal reasoning for large language models in contextualized clinical scenarios of laboratory test interpretation. *NPJ Digit Med*. 2026 Apr 23. doi: 10.1038/s41746-026-02632-3. Epub ahead of print. PMID: 42020503.
3. Bhasuran, B., Abdulkadhar, S., & Natarajan, J. (2026). Unraveling the Biomarker Prospects of High-Altitude Diseases: Insights from Biomolecular Event Network Constructed Using Text Mining. *Medinformatics*.
<https://doi.org/10.47852/bonviewMEDIN62027833>
4. Bhasuran B, Jin Q, Xie Y, Yang C, Hanna K, Costa J, Shavor C, Han W, Lu Z, He Z. Preliminary analysis of the impact of lab results on large language model generated differential diagnoses. *npj Digital Medicine*. 2025 Mar 18;8(1):166.
5. Guillot J, Williams CYK, Azzam S, Bhasuran B, Fernandes G, Ru B, Yang J, Zhang X, Shankar RR, Ge J, Rudrapatna VA. Risk Prediction in Patients With Metabolic Dysfunction-Associated Steatohepatitis Using Natural Language Processing. *Gastro Hep Adv*. 2025 May 14;4(9):100701. doi: 10.1016/j.gastha.2025.100701. PMID: 40688387; PMCID: PMC12271788.
6. X. Wang *et al.*, "Lab-Ai: Using Retrieval Augmentation to Enhance Language Models for Personalized Lab Test Interpretation in Clinical Medicine," *2025 IEEE 13th International Conference on Healthcare Informatics (ICHI)*, Rende, Italy, 2025, pp. 359-368, doi: 10.1109/ICHI64645.2025.00025.
7. Han W, Bhasuran B, Muse VP, Brunak S, Lin L, Hanna K, Huang Y, Bian J, He Z. Assessing the Seasonality of Lab Tests Among Patients with Alzheimer's Disease and Related Dementias in OneFlorida Data Trust. *AMIA Annu Symp Proc*. 2025 May 22;2024:483-492. PMID: 40417579; PMCID: PMC12099435.

8. Manoharan, S., Iyappan, O. R., Prabahar, A., Bhasuran, B., & Raja, K. (2025). Gut-lung axis microbiome: Towards precision medicine in respiratory disorders. *World Academy of Sciences Journal*, 7(5), 1-15.
9. Oviya, Iyappan Ramalakshmi, Guruprasath Manika Rameshbabu, Shree Prasad Muthukrishnan, Balu Bhasuran, Tharun Kaarthik Gunasekaran Kumutha, Sudeesh Kumar Venkatesh, and Shalini Deena Dhayalan. "Identifying lncRNA–Disease Association Based on Dormant Factor Model." *Medinformatics* (2025).
10. Bhasuran, B., Murugesan, G., & Natarajan, J. (2025). Literature-Based Discovery (LBD): Towards Hypothesis Generation and Knowledge Discovery in Biomedical Text Mining. *Medinformatics*, 2(4), 241–255.
<https://doi.org/10.47852/bonviewMEDIN52025348>
11. I. R. Oviya *et al.*, "GRU-CNN Hybrid Model Approach to Classify Human Brain Emotion using EEG Dataset," *2025 Third International Conference on Networks, Multimedia and Information Technology (NMITCON)*, BENGALURU, India, 2025, pp. 1-5, doi: 10.1109/NMITCON65824.2025.11188262.
12. Bhasuran B, Schmolly K, Kapoor Y, Jayakumar NL, Doan R, Amin J, Meninger S, Cheng N, Deering R, Anderson K, Beaven SW, Wang B, Rudrapatna VA. Reducing diagnostic delays in acute hepatic porphyria using health records data and machine learning. *J Am Med Inform Assoc*. 2024 Jul 1;ocae141. doi: 10.1093/jamia/ocae141. Epub ahead of print. PMID: 38946554.
13. He Z, Bhasuran B, Jin Q, Tian S, Hanna K, Shavor C, Arguello LG, Murray P, Lu Z. Quality of Answers of Generative Large Language Models Versus Peer Users for Interpreting Laboratory Test Results for Lay Patients: Evaluation Study. *J Med Internet Res*. 2024 Apr 17;26:e56655. doi: 10.2196/56655. PMID: 38630520; PMCID: PMC11063893.
14. Bhasuran, Balu, et al. "Predicting HIV Diagnosis Among Emerging Adults Using Electronic Health Records and Health Survey Data in All of Us Research Program." 2024 IEEE International Conference on Bioinformatics and Biomedicine (BIBM). IEEE, 2024.
15. Silverman AL, Bhasuran B, Mosenia A, Yasini F, Ramasamy G, Banerjee I, Gupta S, Mardrossian T, Narain R, Sewell J, Butte AJ, Rudrapatna VA. Accurate, Robust, and Scalable Machine Abstraction of Mayo Endoscopic Subscores From Colonoscopy Reports. *Inflamm Bowel Dis*. 2025 Mar 3;31(3):665-670. doi: 10.1093/ibd/izae068. PMID: 38533919; PMCID: PMC11879245.
16. Silverman AL, Sushil M, Bhasuran B, Ludwig D, Buchanan J, Racz R, Parakala M, El-Kamary S, Ahima O, Belov A, Choi L, Billings M, Li Y, Habal N, Liu Q, Tiwari J, Butte AJ, Rudrapatna VA. Algorithmic Identification of Treatment-Emergent Adverse Events From Clinical Notes Using Large Language Models: A Pilot Study in Inflammatory Bowel Disease. *Clin Pharmacol Ther*. 2024 Jun;115(6):1391-1399. doi: 10.1002/cpt.3226. Epub 2024 Mar 8. PMID: 38459719; PMCID: PMC11090709.
17. Patel PV, Zhang A, Bhasuran B, Ravindranath VG, Heyman MB, Verstraete SG, Butte AJ, Rosen MJ, Rudrapatna VA; ImproveCareNow Pediatric IBD Learning Health System. Real-world effectiveness of ustekinumab and vedolizumab in TNF-exposed pediatric patients with ulcerative colitis. *J Pediatr Gastroenterol Nutr*. 2024 May;78(5):1126-1134. doi: 10.1002/jpn3.12169. Epub 2024 Mar 14. PMID: 38482890; PMCID: PMC11065561.
18. Bhasuran B, Manoharan S, Iyyappan OR, Murugesan G, Prabahar A, Raja K. Large Language Models and Genomics for Summarizing the Role of microRNA in Regulating mRNA Expression. *Biomedicines*. 2024 Jul 10;12(7):1535. doi: 10.3390/biomedicines12071535. PMID: 39062108; PMCID: PMC11274411.
19. Yadav, A., Balu Bhasuran & Oviya, I. R. (2023, November). A Novel Approach for Classifying DNA Barcodes Using Ensemble NLP Models. In 2023 International Conference on Research Methodologies in Knowledge Management, Artificial Intelligence and Telecommunication Engineering (RMKMATE) (pp. 1-5). IEEE.
20. Kunchapu, Abhiram, I. R. Oviya, and Balu Bhasuran. "Precision Enhanced Breast Cancer Prediction Using Deep Learning Models." In 2023 International Conference on Artificial Intelligence for Innovations in Healthcare Industries (ICAIHI), vol. 1, pp. 1-6. IEEE, 2023.
21. Balu Bhasuran*, and Jeyakumar Natarajan. "DisGeReExT: a knowledge discovery system for exploration of disease–gene associations through large-scale literature-wide analysis study." *Knowledge and Information Systems* (2023): 1-25
22. Balu Bhasuran*. BioBERT and Similar Approaches for Relation Extraction. *Methods Mol Biol*. 2022;2496:221-235. doi: 10.1007/978-1-0716-2305-3_12. PMID: 35713867.
23. Balu Bhasuran*. Combining Literature Mining and Machine Learning for Predicting Biomedical Discoveries. *Methods Mol Biol*. 2022;2496:123-140. doi: 10.1007/978-1-0716-2305-3_7. PMID: 35713862.
24. Natarajan, Jeyakumar, Balu Bhasuran*, and Gurusamy Murugesan. "Big Data Analytics: A Text Mining Perspective and Applications in Biomedicine and Healthcare." In *Big Data Applications in Industry 4.0*, pp. 367-408. Auerbach Publications, 2022.

25. Maroli N, Balu Bhasuran, Natarajan J, Kolandaivel P. The potential role of procyanidin as a therapeutic agent against SARS-CoV-2: a text mining, molecular docking and molecular dynamics simulation approach. *J Biomol Struct Dyn*. 2022 Feb;40(3):1230-1245. doi: 10.1080/07391102.2020.1823887. Epub 2020 Sep 22. PMID: 32960159; PMCID: PMC7544928.
26. Sabenabanu Abdulkadhar, Balu Bhasuran, and Jeyakumar Natarajan. "Multiscale laplacian graph kernel combined with lexico-syntactic patterns for biomedical event extraction from literature." *Knowledge and Information Systems* (2020) 1-31
27. Maroli N, Kalagatur NK, Balu Bhasuran, Jayakrishnan A, Manoharan RR, Kolandaivel P, Natarajan J, Kadirvelu K. Molecular Mechanism of T-2 Toxin-Induced Cerebral Edema by Aquaporin-4 Blocking and Permeation. *J Chem Inf Model*. 2019 Nov 25;59(11):4942-4958. doi: 10.1021/acs.jcim.9b00711. Epub 2019 Nov 5. PMID: 31644276.
28. Balu Bhasuran & Natarajan, J. (2019). Distant supervision for large-scale extraction of gene–disease associations from literature using DeepDive. In *International Conference on Innovative Computing and Communications: Proceedings of ICICC 2018*, Volume 2 (pp. 367-374). Springer Singapore.
29. Subramanian D, Balu Bhasuran, Natarajan J. Genomic analysis of RNA-Seq and sRNA-Seq data identifies potential regulatory sRNAs and their functional roles in *Staphylococcus aureus*. *Genomics*. 2019 Dec;111(6):1431-1446. doi: 10.1016/j.ygeno.2018.09.016. Epub 2018 Oct 7. PMID: 30304708.
30. Balu Bhasuran, Natarajan J. Automatic extraction of gene-disease associations from literature using joint ensemble learning. *PLoS One*. 2018 Jul 26;13(7):e0200699. doi: 10.1371/journal.pone.0200699. PMID: 30048465; PMCID: PMC6061985.
31. Balu Bhasuran*, Subramanian D, Natarajan J. Text mining and network analysis to find functional associations of genes in high altitude diseases. *Comput Biol Chem*. 2018 Aug;75:101-110. doi: 10.1016/j.compbiolchem.2018.05.002. Epub 2018 May 2. PMID: 29763853.
32. Murugesan G, Abdulkadhar S, Balu Bhasuran, Natarajan J. BCC-NER: bidirectional, contextual clues named entity tagger for gene/protein mention recognition. *EURASIP J Bioinform Syst Biol*. 2017 Dec;2017(1):7. doi: 10.1186/s13637-017-0060-6. Epub 2017 May 5. PMID: 28477208; PMCID: PMC5419958.
33. Balu Bhasuran, Murugesan G, Abdulkadhar S, Natarajan J. Stacked ensemble combined with fuzzy matching for biomedical named entity recognition of diseases. *J Biomed Inform*. 2016 Dec;64:1-9. doi: 10.1016/j.jbi.2016.09.009. Epub 2016 Sep 12. PMID: 27634494.

PREPRINTS

34. Bhasuran B, Jin Q, Deville A, Wu Y, Hanna K, Lu Z, He Z. LabQAR: A Manually Curated Dataset for Question Answering on Laboratory Test Reference Ranges and Interpretation. *medRxiv [Preprint]*. 2025 Jun 3:2025.06.03.25328882. doi: 10.1101/2025.06.03.25328882. PMID: 40502613; PMCID: PMC12155050. (Under review in Scientific Data (Nature))
35. George, Ajo Babu, Anna Mariam John, Athul Anoop, and Balu Bhasuran. "A Unified Multimodal Framework for Dataset Construction and Model-Based Diagnosis of Ameloblastoma." *arXiv preprint arXiv:2602.05515* (2026).
36. Manoharan, Sharanya, Balu Bhasuran, Oviya Ramalakshmi Iyyappan, Mohamed Saleem Abdul Shukkoor, Malathi Sellapan, and Kalpana Raja. "AI-Driven Drug Repurposing through miRNA-mRNA Relation." *arXiv preprint arXiv:2509.03336* (2025).
37. Bhasuran B, Wang X, Gupta D, Killian M, He Z. Predicting Organ Rejections for Pediatric Heart Transplantations with a Combined Use of Transplant Registry Data and Electronic Health Records. *medRxiv [Preprint]*. 2025 May 1:2025.04.29.25326701. doi: 10.1101/2025.04.29.25326701. PMID: 40343033; PMCID: PMC12060946.

CONFERENCE ABSTRACTS

38. He Z, Bhasuran B, Jin Q, Tian S, Hanna K, Lu Z. USING INFORMATICS AND GENERATIVE AI TO SUPPORT OLDER ADULTS' UNDERSTANDING OF LAB TEST RESULTS. *Innovation in Aging*. 2024 Dec 31;8(Suppl 1):109.
39. Balu Bhasuran, Shadera Slatter, Gail Fernandes, Boshu Ru, Joe Yang, Xiao Zhang, Ravi Shankar, Jin Ge, and Vivek Rudrapatna. NASHDetection: A Natural Language Processing Method for Identifying Patients With Non-Alcoholic Steatohepatitis Using Clinical Notes. *Official journal of the American College of Gastroenterology | ACG*, 2023.
40. Balu Bhasuran, Shadera Slatter, Gail Fernandes, Boshu Ru, Joe Yang, Xiao Zhang, Ravi Shankar, Jin Ge, and Vivek Rudrapatna. Uncontrolled Diabetes and Hypertension Are Associated With the Risk of New-Onset Cirrhosis in Patients With Nonalcoholic Steatohepatitis *Official journal of the American College of Gastroenterology | ACG*, 2023.
41. Anna L Silverman, Madhumita Sushil, Balu Bhasuran, Dana Ludwig, James Buchanan, Rebecca Racz, Mahalakshmi Parakala, Samer El-Kamary, Ohenewaa Ahima, Artur Belov, Lauren Choi, Monisha Billings, Yan Li, Nadia Habal, Qi Liu, Jawahar Tiwari,

Atul Butte, and Vivek Rudrapatna. Algorithmic identification of treatment-emergent adverse events from clinical notes using large language models: A pilot study in inflammatory bowel disease. Official journal of the American College of Gastroenterology| ACG, 2023.

42. Silverman, Anna L., Balu Bhasuran, Arman Mosenia, Fatema Yasini, Saransh Gupta, Narbe Mardirossian, Rohan Narain, Justin L. Sewell, Atul Butte, and Vivek A. Rudrapatna. "Accurate, Robust, And Scalable Abstraction Of Mayo Endoscopic Subscores From Colonoscopy Reports." In GASTROENTEROLOGY, vol. 162, no. 7, pp. S618-S619. 1600 John F Kennedy Boulevard, Ste 1800, Philadelphia, Pa 19103-2899 Usa: Wb Saunders Co-Elsevier Inc, 2022.
-